



Institute of
Information Technology

CYBERSECURITY LABORATORY

Lab Work Report

Vulnerability Analysis

Integrated Report for Lab Work 01–04

Intrusion Detection System and Intrusion Prevention System

All Slide Shows: <https://cygnussec.com/project/ids-ips>



Report Requirements – Scope

Select and analyze a vulnerability in **Metasploitable2** or another approved vulnerable target.

- The target must be part of the authorized laboratory environment.
- The report must connect discovery, validation, exploitation, detection, and prevention.
- Every result must include evidence and an explanation.



Report Requirements – Workflow

1. Discover and assess the vulnerability using **Nmap**, **NSE** and **Greenbone/OpenVAS**.
2. Explain the vulnerability and provide evidence from the scan results.
3. Exploit it using the **Metasploit Framework** or another appropriate tool.
4. Propose and test detection and prevention controls.



Report Outline I – Vulnerability Introduction

1. Overview

- What is the vulnerability?
- What category or type of vulnerability is it?

2. Technical mechanism

- Explain how the vulnerability works internally.

3. Impact and severity

- Describe the potential impact.
- Discuss the severity using CVSS or another recognized metric.



Report Outline II – Test Environment

Create and document an isolated environment for testing.

- Identify the attacker, firewall/IDS, and victim systems.
- Record network interfaces, IP addresses, and routing.
- Verify connectivity before scanning.
- Explain why the environment is safe and authorized.



Report Outline II – Vulnerability Scanning

- Use **Nmap** for scanning steps 1–3 and capture the results.
- Use **OpenVAS/Greenbone** for scanning step 4.
- Create and document the scan target and task.
- Run the scan, export the report, and capture evidence of the selected vulnerability.



Report Outline II – Exploitation

Document the implementation in three stages:

1. Create the authorized testing environment.
2. Scan and validate the selected vulnerability.
3. Exploit the vulnerability using:
 - **Metasploit Framework**, or
 - another justified exploitation tool or technique.

Include commands, relevant configuration, screenshots, and an explanation of each result.



Mitigation and Remediation

- **Detection**: use **Snort** or another IDS/IPS tool from Lab Work 04.
- **Prevention**: apply firewall controls from Lab Work 03.
- Explain how each control detects, blocks, or reduces the vulnerability's impact.
- Provide evidence that the mitigation or remediation works.



Conclusion and References

- Summarize the vulnerability, exploitation result, and defensive controls.
- Reiterate why the vulnerability must be addressed.
- State any limitations and recommended next steps.
- List all sources, tools, advisories, CVE/CVSS records, and vulnerability databases used.



Submission Instructions

Deadline: 23:59, November 1, 2026

- File title: Student-ID_Full-Name
- Example: BI13-001_Nguyen-Van-A
- Submit by email to: duongtd@cygnussec.tech



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Thank You!

Questions and Discussion